MTH 200 Calculating Derivatives (up through 3.5) & Definition of the Derivative

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. You may recall that we recently learned the derivative of $y=\sec(x)$.

a) Without referencing your notes or other sources attempt to prove/derive the derivative rule for $y=\sec(x)$.

We note that $y=\sec(x)=\dfrac{1}{\cos(x)}$.

$$\frac{dy}{dx}=\frac{\cos(x)\frac{d}{dx}[1]-1\frac{d}{dx}\left[\cos(x)\right]}{\cos^2(x)}=\frac{\cos(x)\cdot 0-1\left[-\sin(x)\right]}{\cos^2(x)}=\frac{\sin(x)}{\cos^2(x)}$$
$$=\frac{1}{\cos(x)}\cdot\frac{\sin(x)}{\cos(x)}$$
$$=\sec(x)\tan(x)$$

<u>Summary Point</u>

b) What are some of the benefits of knowing how to derive a derivative rule versus just having it memorized? What are some of the benefits of having it memorized versus always having to derive it?

If you are able to derive a derivative rule you understand more about where it comes from and it can make it easier to memorize. You can always re-derive the rule if you happen to forget it.

If you can only derive a rule but don't have it memorized then you will always take way longer to find derivatives. Some things are worth dedicating to memory.

2. Show that if $f(x)=\dfrac{\sin(x)}{1+\cos(x)}$ then $f'(x)=\dfrac{1}{1+\cos(x)}$.

$$f'(x)=\frac{(1+\cos(x))\cos(x)-(-\sin(x))\sin(x)}{(1+\cos(x))^2}=\frac{\cos(x)+\cos^2(x)+\sin^2(x)}{(1+\cos(x))^2}=\frac{\cos(x)+1}{(1+\cos(x))^2}=\frac{1}{1+\cos(x)}$$

3. Determine the equation of a line which is perpendicular to $y=\dfrac{\tan(x)}{1+\tan(x)}$ at $x=\dfrac{\pi}{4}$.

We first note that $y'=\dfrac{(1+\tan(x))\sec^2(x)-\sec^2(x)\tan(x)}{(1+\tan(x))^2}$ and

$$y'\left(\frac{\pi}{4}\right)=\frac{\left(1+\tan\left(\frac{\pi}{4}\right)\right)\sec^2\left(\frac{\pi}{4}\right)-\sec^2\left(\frac{\pi}{4}\right)\tan\left(\frac{\pi}{4}\right)}{\left(1+\tan\left(\frac{\pi}{4}\right)\right)^2}=\frac{(1+1)(2)-(2)(1)}{(1+1)^2}=\frac{2}{4}=\frac{1}{2}$$ and

$$y\left(\frac{\pi}{4}\right)=\frac{\tan\left(\frac{\pi}{4}\right)}{1+\tan\left(\frac{\pi}{4}\right)}=\frac{1}{1+1}=\frac{1}{2}$$

Therefore, for this tangent line we have

$m=-2$

Point: $\left(\dfrac{\pi}{4},\dfrac{1}{2}\right)$

So the equation for the tangent line is: $y-\dfrac{1}{2}=-2\left(x-\dfrac{\pi}{4}\right)\ \rightarrow\ y=-2x+\dfrac{\pi+1}{2}$

4. Suppose that $f(x)=\csc(x)$. For what values of x will the slopes of tangent lines on $y=f'(x)$ be horizontal on $[0,2\pi]$.

To find the slopes of tangent lines on $f'(x)$ we need to find $f''(x)$.
Note that

$$f'(x)=-\csc(x)\cot(x) \text{ and}$$
$$f''(x)=\csc(x)\csc^2(x)+\csc(x)\cot(x)\cot(x)=\csc^3(x)-\csc(x)\cot^2(x)$$

We know that $f''(x)=0$ when

$$\csc^3(x)-\csc(x)\cot^2(x)=0 \quad\rightarrow\quad \csc(x)\left[\csc^2(x)-\cot^2(x)\right]=0$$

This implies that $\csc(x)=0$ *or* $\csc^2(x)-\cot^2(x)=0$

$\csc(x)$ never equals 0.

$$\csc^2(x)-\cot^2(x)=0 \quad\rightarrow\quad \csc^2(x)=\cot^2(x) \quad\rightarrow\quad 1=\cos^2(x) \quad\rightarrow\quad \cos(x)=\pm 1 \quad\rightarrow\quad x=\frac{\pi}{2},\frac{3\pi}{2}.$$

5. Evaluate $\lim\limits_{h\to 0}\dfrac{\sin\left(\frac{\pi}{3}+h\right)-\sin\left(\frac{\pi}{3}\right)}{h}$.

We recognize this to be the derivative of $y=\sin(x)$ evaluated at $\frac{\pi}{3}$.

$$\lim_{h\to 0}\frac{\sin\left(\frac{\pi}{3}+h\right)-\sin\left(\frac{\pi}{3}\right)}{h}=\frac{d}{dx}\left[\sin(x)\right]\Big|_{x=\frac{\pi}{3}}=\cos\left(\frac{\pi}{3}\right)=\frac{1}{2}$$

6. Determine all values of x at which the slopes of tangent lines on $f(x)=\sin(x)$ and $g(x)=\cos(x)$ are opposite values.

Note that $f'(x)=\cos(x)$ and $g'(x)=-\sin(x)$.
The slopes of tangent lines have opposite values for all values of x where
$\cos(x)=\sin(x) \quad\rightarrow\quad \tan(x)=1$.

Therefore, $x=\frac{\pi}{4}+\pi k$, where k is an integer.

Summary Point

7. Show that the function $f(x)=\begin{cases} x^2\cos\left(\dfrac{1}{x}\right) & if\ \ x\neq 0 \\ 0 & if\ \ x=0 \end{cases}$ is differentiable at $x=0$. *Hint:* You need to use the definition of the derivative.

We know that in order for a function to be differentiable at $x=a$ it must be the case that $f'(a)$ exists, that is it must be true that the limit $\lim\limits_{h\to 0}\dfrac{f(a+h)-f(a)}{h}$ exists.

Since we want to show that $f(x)$ is differentiable at $x=0$ we must show that $\lim\limits_{h\to 0}\dfrac{f(0+h)-f(0)}{h}$ exists.

We see that $\lim\limits_{h\to 0}\dfrac{f(0+h)-f(0)}{h}=\lim\limits_{h\to 0}\dfrac{f(h)-0}{h}=\lim\limits_{h\to 0}\dfrac{f(h)}{h}=\lim\limits_{h\to 0}\dfrac{h^2\cos\left(\frac{1}{h}\right)}{h}=\lim\limits_{h\to 0}h\cos\left(\dfrac{1}{h}\right)$.

We note that for any non-zero value of h that $-1\leq\cos\left(\dfrac{1}{h}\right)\leq 1$. Then,

if $h>0$, $\quad -h\leq h\cos\left(\dfrac{1}{h}\right)\leq h$ and

if $h<0$, $\quad -h\geq h\cos\left(\dfrac{1}{h}\right)\geq h$ and

In either case we still have that $\lim\limits_{h\to 0}h=0$ & $\lim\limits_{h\to 0}(-h)=0$ which implies, by the Squeeze Theorem, that $\lim\limits_{h\to 0}h\cos\left(\dfrac{1}{h}\right)=0$. Therefore, we have showed that $\lim\limits_{h\to 0}\dfrac{f(0+h)-f(0)}{h}=0$ and thus exists. So, $f(x)$ is differentiable at $x=0$.